

# Juan Diaz-Colunga, Ph.D.



Junior Group Leader – Microbial Systems Biology Lab  
Dairy Research Institute of Asturias (IPLA)  
Spanish National Research Council (CSIC)

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I am a trained physicist with an M.Sc. in Biomedical Engineering and a Ph.D. in Biophysics. I currently lead the Microbial Systems Biology laboratory ([MSBlab](#)) at the Dairy Research Institute of Asturias (IPLA-CSIC) in Oviedo, Spain, after earning a “Ramon y Cajal” fellowship with the top-2 score in the area of Biosciences and Biotechnology (call of 2023). I have secured over 500,000€ in funding since the establishment of the lab in early 2025, which are supporting my research in the areas of quantitative microbial ecology & evolution. My approach combines theory, computational modeling, and wet-lab experimentation.

|                               |                                                                                                                      |             |
|-------------------------------|----------------------------------------------------------------------------------------------------------------------|-------------|
| <b>Keywords</b>               | Biophysics, Systems Biology, Ecology & Evolution, Population Genetics                                                |             |
| <b>Education</b>              | <b>Ph.D. Biophysics</b> , Spanish National Center for Biotechnology (CNB)                                            | 2015 – 2019 |
|                               | <b>M.Sc. Biomedical Engineering</b> , Universidad Politecnica de Madrid                                              | 2013 – 2014 |
|                               | <b>B.Sc. Physics</b> , Universidad Autonoma de Madrid                                                                | 2009 – 2013 |
| <b>Awards and fellowships</b> | <b>“Ramon y Cajal” Fellowship</b> , top-2 candidate in Biosciences & Biotechnology                                   | 2024        |
|                               | <b>PhD cum laude</b> with an <i>international mention</i>                                                            | 2019        |
|                               | <b>“Severo Ochoa” PhD Fellowship</b>                                                                                 | 2015 – 2019 |
|                               | <b>Undergraduate Excellence Fellowship</b> , Comunidad de Madrid                                                     | 2009 – 2013 |
| <b>Positions</b>              | <b>Junior Group Leader</b> , Dairy Research Institute of Asturias (IPLA-CSIC)                                        | 2025 – now  |
|                               | <b>Postdoctoral Scientist</b>                                                                                        |             |
|                               | University of Salamanca, Institute of Functional Biology & Genomics                                                  | 2023 – 2024 |
|                               | Yale University, Dept. of Ecology & Evolutionary Biology                                                             | 2020 – 2022 |
|                               | Universidad Autonoma de Madrid, Dept. of Biochemistry                                                                | 2019 – 2020 |
|                               | <b>Visiting Scholar</b> , Massachusetts Institute of Technology (MIT)                                                | 2018        |
|                               | <b>PhD Fellow</b> , Spanish National Center for Biotechnology (CNB-CSIC)                                             | 2015 – 2019 |
| <b>Grants</b>                 | <b>Spanish National Research Plan</b> (ref. PID2024-162835NA-I00)                                                    | 2025 – 2028 |
|                               | Title: <i>Genome Evolution with Global Epistasis (GENESIS)</i>                                                       |             |
|                               | Awarded by the Spanish National Research Agency (AEI)                                                                |             |
|                               | 137,500€ (direct funding) + 134,200€ (PhD contract)                                                                  |             |
|                               | <b>MaX-CSIC Excellence Grant</b> (ref. 20252MAX002)                                                                  | 2024 – 2028 |
|                               | Title: <i>Functional dynamics in microbial systems: from molecular interactions to their design and optimization</i> |             |
|                               | Awarded by the Spanish National Research Council (CSIC)                                                              |             |
|                               | 300,000€                                                                                                             |             |
|                               | <b>“Ramon y Cajal” research aid</b> (ref. RYC2023-045580-I)                                                          | 2024 – 2029 |
|                               | Awarded by the Spanish National Research Agency (AEI)                                                                |             |
|                               | 50,000€                                                                                                              |             |

## Publications

Metrics: 20 papers (since 2016), 1166 citations, h-index = 18 (Google Scholar, checked on August 2026)

\* (co-)first author    ✉ (co-)corresponding author

1. Andrea Arrabal, Magdalena San Roman, Juan Diaz-Colunga, Alvaro Sanchez. Statistical learning of bacterial growth in combinatorially constructed environments. **bioRxiv** 2026.06.30.735472 (2026)
2. Juan Diaz-Colunga \*, Pablo Catalan, Magdalena San Roman, Andrea Arrabal, Alvaro Sanchez. Full factorial construction of synthetic microbial communities. **eLife** 13:RP101906 (2026)
3. Magdalena San Roman, Andrea Arrabal, Belen Benitez-Dominguez, Isabel Quiros-Rodriguez, Juan Diaz-Colunga ✉. Towards synthetic ecology: strategies for the optimization of microbial community functions. **Frontiers in Synthetic Biology** 3:1532846 (2025)
4. Juan Diaz-Colunga \* ✉, Abigail Skwara, Jean CC Vila, Djordje Bajic, Álvaro Sánchez. Global epistasis and the emergence of function in microbial consortia. **Cell** 187:1–12 (2024)
5. Javier DelaFuente, Juan Diaz-Colunga \*, Alvaro Sanchez, Alvaro San Millan. Global epistasis in plasmid-mediated antimicrobial resistance. **Molecular Systems Biology** 20(4):311–20 (2024)
6. Álvaro Sánchez, Andrea Arrabal, Magdalena San Román, Juan Diaz-Colunga. The optimization of microbial functions through rational environmental manipulations. **Molecular Microbiology** 00:1–10 (2024)
7. Juan Diaz-Colunga \* ✉, Alvaro Sanchez, C. Brandon Ogbunugafor. Environmental modulation of global epistasis in a drug resistance fitness landscape. **Nature Communications** 14:8055 (2023)
8. Abigail Skwara, Karna Gowda, Mahmoud Yousef, Juan Diaz-Colunga, Arjun S Raman, Alvaro Sanchez, Mikhail Tikhonov, Seppe Kuehn. Statistically learning the functional landscape of microbial communities. **Nature Ecology & Evolution** 7:1823–1833 (2023)
9. Juan Diaz-Colunga \* ✉, Abigail Skwara, Karna Gowda, Ramon Diaz-Uriarte, Mikhail Tikhonov, Djordje Bajic, Alvaro Sanchez. Global epistasis on fitness landscapes. **Philosophical Transactions of the Royal Society B** 378:20220053 (2023)
10. Alvaro Sanchez, Djordje Bajic, Juan Diaz-Colunga \*, Abigail Skwara, Jean CC Vila, Seppe Kuehn. The community-function landscape of microbial consortia. **Cell Systems** 14(2):122–34 (2023)
11. Javier Ruiz, Miguel de Celis, Juan Diaz-Colunga, Jean CC Vila, Belen Benitez-Dominguez, Javier Vicente, Antonio Santos, Alvaro Sanchez, Ignacio Belda. Predictability of the community-function landscape in wine yeast ecosystems. **Molecular Systems Biology** 19(9):e11613 (2023)
12. Juan Diaz-Colunga \*, Nanxi Lu, Alicia Sanchez-Gorostiaga, Chang-Yu Chang, Helen S Cai, Joshua E Goldford, Mikhail Tikhonov, Álvaro Sánchez. Top-down and bottom-up cohesiveness in microbial community coalescence. **PNAS** 119(6):e2111261119 (2022)
13. Sylvie Estrela, Juan Diaz-Colunga \*, Jean CC Vila, Alicia Sanchez-Gorostiaga, Alvaro Sanchez. Diversity begets diversity under microbial niche construction. **eLife** (accepted) (2022)
14. Álvaro Sánchez, Jean CC Vila, Chang-Yu Chang, Juan Diaz-Colunga, Sylvie Estrela, María Rebolleda-Gomez. Directed evolution of microbial communities. **Annual Review of Biophysics** 50:323–41 (2021)
15. Chang-Yu Chang, Jean CC Vila, [...], Juan Diaz-Colunga, Sylvie Estrela, Maria Rebolleda-Gomez, Alvaro Sanchez. Engineering complex communities by directed evolution. **Nature Ecology & Evolution** 5(7):1011–23 (2021)
16. Juan Diaz-Colunga \*, Ramon Diaz-Uriarte. Conditional prediction of consecutive tumor evolution using cancer progression models: What genotype comes next? **PLOS Computational Biology** 17(12):e1009055 (2021)
17. Ana F Lima, Gillian May, Juan Diaz-Colunga, Susana Pedreiro, Artur Paiva, Luciana Ferreira, Tariq Enver, Francisco J Iborra, Ricardo Pires das Neves. Osmotic modulation of chromatin impacts on efficiency and kinetics of cell fate modulation. **Scientific Reports** 8(1):1–14 (2018)
18. Silvia Márquez-Jurado, Juan Diaz-Colunga \*, Ricardo Pires das Neves, Antonio Martinez-Lorente, Fernando Almazán, Raúl Guantes, Francisco J Iborra. Mitochondrial levels determine variability in cell death by modulating apoptotic gene expression. **Nature Communications** 9(1):1–11 (2018)
19. Laura Marcos-Villar, Juan Diaz-Colunga, Juan Sandoval, Noelia Zamarreño, Sara Landeras-Bueno, Manel Esteller, Ana Falcón, Amelia Nieto. Epigenetic control of influenza virus: role of H3K79 methylation in interferon-induced antiviral response. **Scientific Reports** 8(1):1–13 (2018)
20. Raúl Guantes, Juan Diaz-Colunga, Francisco J Iborra. Mitochondria and the non-genetic origins of cell-to-cell variability: more is different. **BioEssays** 38(1):64–76 (2016)

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|----------------|-----------------------------------------------------------------------------------------|----------------|
| Selected talks | <b>MMEE2026: Mathematical Models in Ecology and Evolution</b>                           | 2026           |
|                | Cork, Ireland                                                                           |                |
|                | <i>Invited talk</i>                                                                     |                |
|                | <b>BiomeFUN: From Genomic Analysis to Functional Models in Microbiomes</b>              | 2025           |
|                | Belgrade, Serbia                                                                        |                |
|                | <i>Invited lecture</i>                                                                  |                |
|                | <b>EPFL Workshop: Evolution by Nature and by Design</b>                                 | 2025           |
|                | École Polytechnique Fédérale de Lausanne, Switzerland                                   |                |
|                | <i>Invited talk</i>                                                                     |                |
|                | <b>EMBO Workshop: Molecular mechanisms in evolution and ecology</b>                     | 2024           |
| Mentoring      | European Molecular Biology Laboratory (EMBL), Heidelberg, Germany                       |                |
|                | <i>Invited talk</i>                                                                     |                |
|                | <b>Simons Foundation Life Sciences Workshop: Function of Evolving Systems</b>           | 2024           |
|                | New York, USA                                                                           |                |
|                | <i>Invited presentation</i>                                                             |                |
|                | <b>Scientific Spring Meeting of the Royal Dutch Society of Microbiology</b>             | 2024           |
|                | Arnhem, the Netherlands                                                                 |                |
|                | <i>Invited talk</i>                                                                     |                |
|                | <b>Industrial Microbiology Seminar Series</b>                                           | 2024           |
|                | Delft University of Technology, the Netherlands                                         |                |
| Reviewing      | <i>Invited talk</i>                                                                     |                |
|                | <b>17th Meeting of the Spanish National Network of Lactic Acid Bacteria</b>             | 2024           |
|                | University of León, Spain                                                               |                |
|                | <i>Invited talk – keynote speaker</i>                                                   |                |
|                | <b>CAB Conference:</b>                                                                  | 2022           |
|                | <b>Microbial Communities at the Interface between Ecology and Evolution</b>             |                |
|                | Mexico City, Mexico                                                                     |                |
|                | <b>XXIX Workshop: Advances in Molecular Biology</b>                                     | 2021           |
|                | Spanish National Center for Biotechnology (CNB-CSIC), virtual seminar                   |                |
|                | <b>PhD Students and Postdocs</b>                                                        |                |
|                | Currently supervising a PhD student (expected graduation in Fall 2029) and two postdocs |                |
|                | <b>Undergraduate students</b>                                                           |                |
|                | B.Sc. Biotechnology, lab practicals, Universidad de Oviedo                              | 2026           |
|                | B.Sc. Biology, Final Research Project, Universidad Complutense de Madrid                | 2023           |
|                | EEB Senior Research (EEB475 & 476), Yale University                                     | 2021           |
|                | <b>Teaching</b>                                                                         |                |
|                | M.Sc. Experimental Methods in Biophysics, Universidad Autonoma de Madrid                | 2019           |
|                | Guest Associate Editor for <i>Genetics</i>                                              | 2026           |
|                | Review Editor for <i>Frontiers in Synthetic Biology</i>                                 | 2024 – present |

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|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| Project Reviewer for the French National Research Agency (ANR)                                                                                                                                                                                                                                                        | 2024           |
| Member of the PhD monitoring committee for students at:                                                                                                                                                                                                                                                               |                |
| University of Lausanne, Switzerland                                                                                                                                                                                                                                                                                   | 2024 – present |
| Université Paris-Saclay, France                                                                                                                                                                                                                                                                                       | 2026 – present |
| Reviewer for <i>Nature Ecology &amp; Evolution</i> , <i>PNAS</i> , <i>Nature Communications</i> , <i>The ISME Journal</i> , <i>eLife</i> , <i>Nature Cities</i> , <i>Philosophical Transactions of the Royal Society B</i> , <i>Global Change Biology</i> , <i>PLOS Computational Biology</i> , among other journals. |                |

## Service

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|-----------------------------------------------------------------------------------------------------------------------------------------------|
| Participant & organizer at various scientific outreach activities (most recent: <i>Pint of Science</i> 2025 & 2026, <i>Science Week</i> 2024) |
| Member of the Training Committee at IPLA-CSIC                                                                                                 |
| Member of the Committee for Gender Equality at IBFG-CSIC                                                                                      |
| Founder of the Commission for Cultural Activities at IBFG-CSIC                                                                                |